

Yakub Mungai

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EDUCATION

• University of Florida

Gainesville, FL

Bachelor of Science in Computer Science (Certificate in AI Fundamentals & Applications)

August 2021 - December 2025

RELEVANT COURSEWORK

- Programming in C++
- Data Structures / Algorithms
- Information & Database Systems
- Programming in Python
- Intro to Software Engineering
- Business Analytics & AI
- Algorithm Abstraction and Design
- Discrete Structures
- AI Fundamentals

TECHNICAL SKILLS

Languages: Python, C++, MATLAB, SQL

Technologies / Frameworks: Git, Visual Studio, PyCharm, pandas, PyGame, SFML, SvelteKit, Tailwind CSS

PROJECTS

Project KOSMOS | *Python, LangChain, OpenAI API*

August 2025 - December 2025

- Developed a Critic Agent using LangChain and OpenAI API to provide continuous mission evaluation and feedback for enhanced fault tolerance.
- Implemented natural language critique analyzing telemetry and maneuvers to detect issues and guide real-time corrective actions.
- Integrated with Kerbal Space Program via kRPC and MechJeb2 for telemetry monitoring and autopilot command assessment.
- Enhanced mission success with multi-agent collaboration and adaptive fault tolerance mechanisms.

Discord AI Headshot Bot | *Python, discord.py, Google Gemini API*

September 2025 - October 2025

- Designed and implemented an asynchronous Discord bot using `discord.py` to handle real-time image uploads and thread-based conversations.
- Integrated Google Gemini AI API for advanced image processing to generate professional headshots with controlled background replacement and attire overlay.
- Managed file I/O operations securely in Python, implementing unique temporary file naming and cleanup routines.
- Leveraged asynchronous programming patterns to maintain responsiveness in multi-user, concurrent Discord interactions.
- Implemented robust error handling, validation, and user feedback loops for seamless UX within Discord threads.

AlexG Automator | *Python, MT5, Pandas, NumPy, SciPy, Matplotlib*

Jan 2026 – Present

- Architected a quantitative system for real-time MT5 data ingestion and multi-timeframe trend detection.
- Developed a fractal market structure algorithm using *SciPy* for swing point identification and ATR filters.
- Designed live *Matplotlib* dashboards to visualize algorithmic decision logic against real-time market data.
- Engineering an execution module for automated order placement via supply/demand and confluence scoring.

Colorstack UF Website Development | *SvelteKit, TailwindCSS*

May 2024 - Aug 2024

- Designed and implemented the Students page, integrating the Google Calendar API to enhance event management and user functionality for our 400+ student members.
- Integrated FullCalendar API to display a dynamic and interactive calendar of events on the website.
- Utilized Tailwind CSS to create a visually appealing and responsive layout, resulting in a 20% reduction in page load times.

EXPERIENCE

Information Technology Support Specialist Intern

May 2021 – June 2021

Sanlam Life Insurance

Dar Es Salaam, Tanzania

- Collaborated with the IT Support Specialist to manage and ensure the security of computing systems for over 30 employees, resulting in a decrease in reported IT issues.
- Conducted regular system assessments to identify and address vulnerabilities promptly, which significantly enhanced overall cybersecurity and system integrity.
- Provided timely and effective resolutions for IT concerns, reducing average issue resolution time by 20%.

INVOLVEMENTS

Student Organizations & Programs, University of Florida

- Member, UF Honors Program
- Recipient, UWC Davis Scholarship
- Vice President, ColorStack UF